

Bioinformatics for engineers

Numerical representation of sequences

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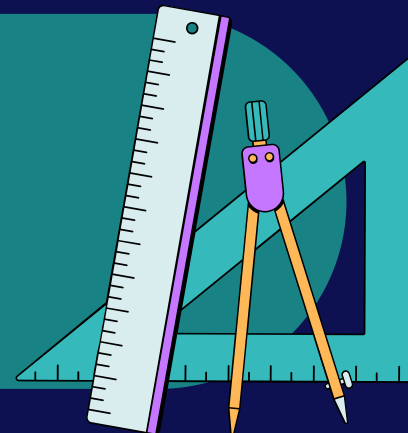
1. Vector encode(string DNA_sequence)



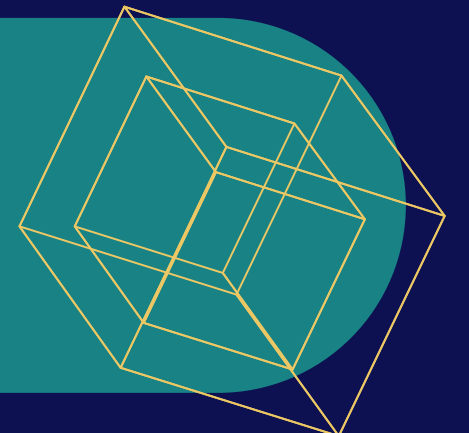
2. Principal Component Analysis



3. How to measure distance?



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SEQUENCE ENCODING & FEATURE EXTRACTION

Sequence Encoding

The problem

Statistical and machine learning algorithms **cannot "read"** text (A, T, C, G). They only operate with *tokens*.

We need the numbers to not just be a name, but to *represent* a biological characteristic or meaning (mass, charge, frequency of occurrence).

The solution

You could assign to each nitrogenous base a number (eg. A=1, T=2, G=3, C=4). *A computer would think that Thymine is "four times more important" than Adenine.*

A solution to this is vector encoding your data. (you can apply this not only to ATGC) So now each nitrogenous base is a *binary vector*, like this:

A = [1, 0, 0, 0]

B = [0, 1, 0, 0]

C = [0, 0, 1, 0]

D = [0, 0, 0, 1]

***This concept is called
one-hot encoding**

Why is this important?

Feature extraction: It allows measuring physicochemical properties or counting k-mers (as repeated patterns)

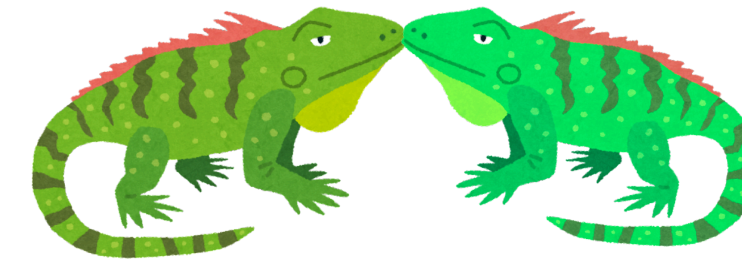
Standardization: Converts sequences of different lengths into fixed-size vectors so they can be compared.

By vectorizing all your samples, you create a Feature Matrix.

PRINCIPAL COMPONENT ANALYSIS (PCA)

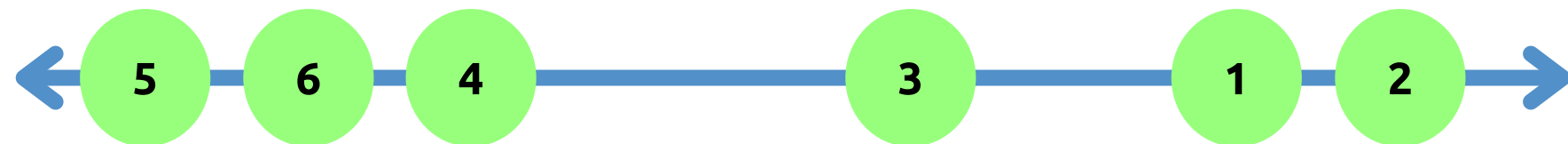
Correlating “variables” and “samples”

	Iguana 1	Iguana 2	Iguana 3	Iguana 4	Iguana 5	Iguana 6
Gene 1	10	11	8	3	1	2
Gene 2	6	4	5	3	2.8	1



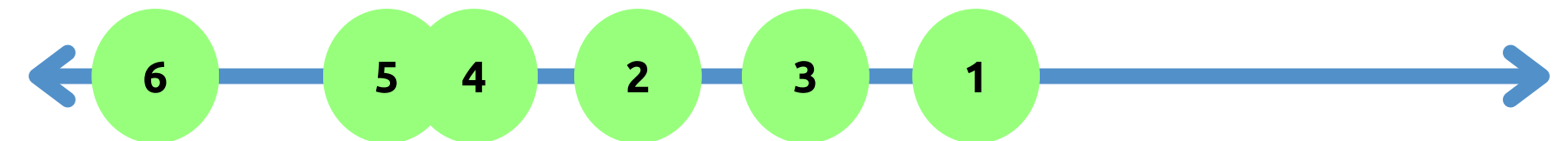
Imagine you quantified the transcription of genes 1 and 2 in 6 different iguanas. You want to see how related are these iguanas to each other.

First, just analyze the first set of data for gene #1



You could tell, that the iguanas 1, 2 and 3 are similar, as well as 4, 5 and 6. There is a difference between both groups

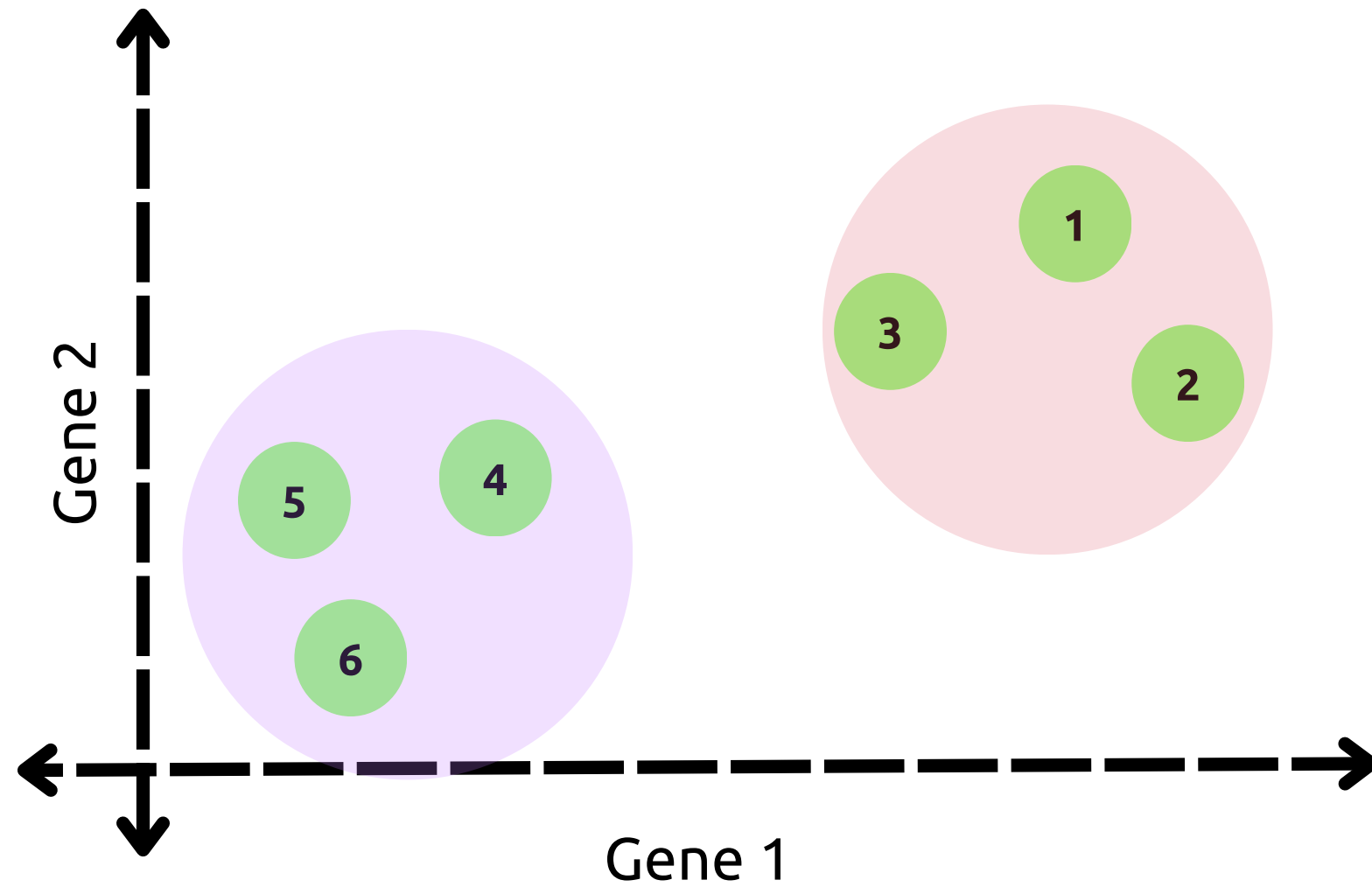
Now, see the same exercise for gene #2



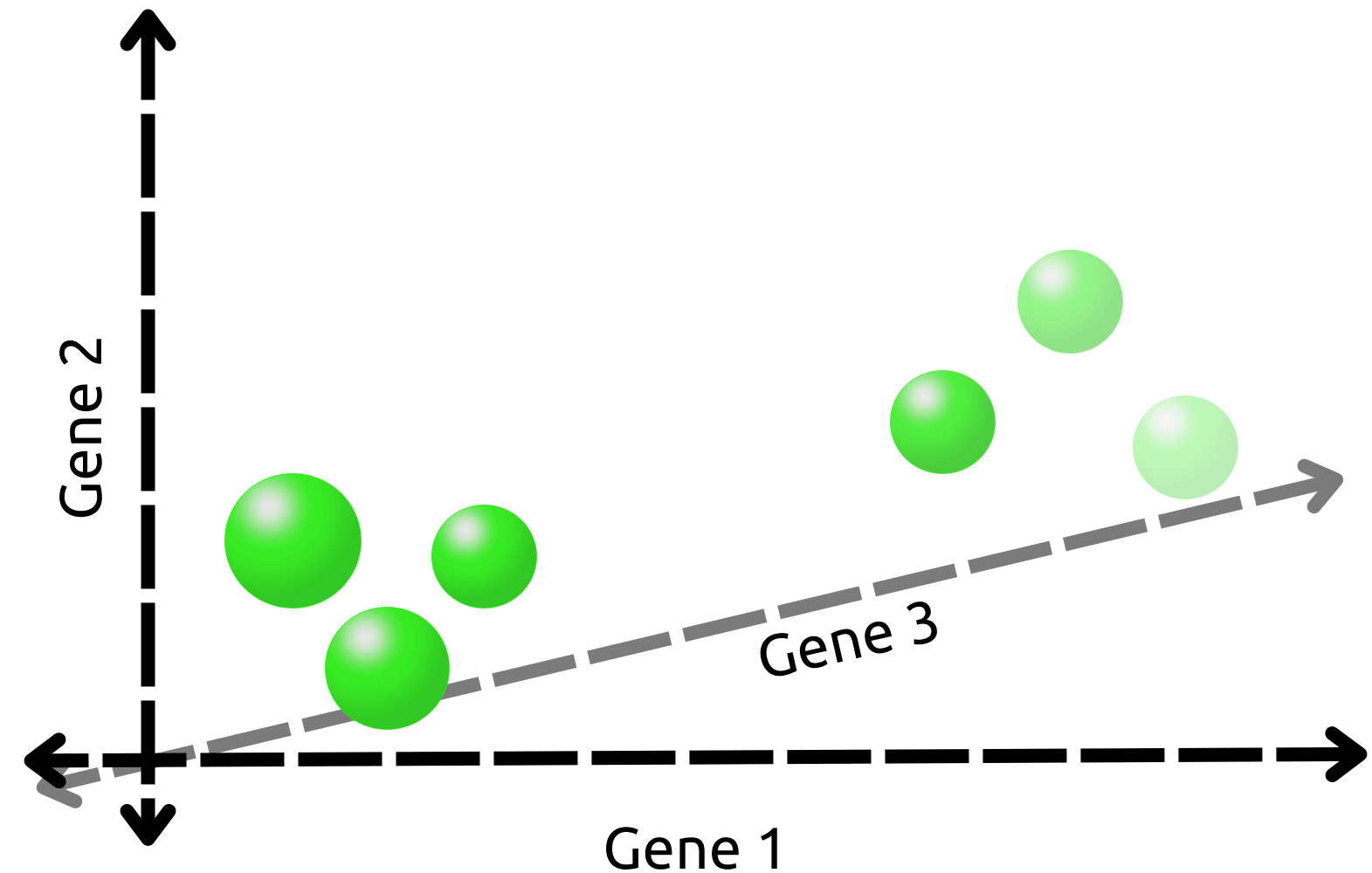
Which iguanas are more similar?

Exactly, this is why we plot...

Correlating “variables” and “samples”



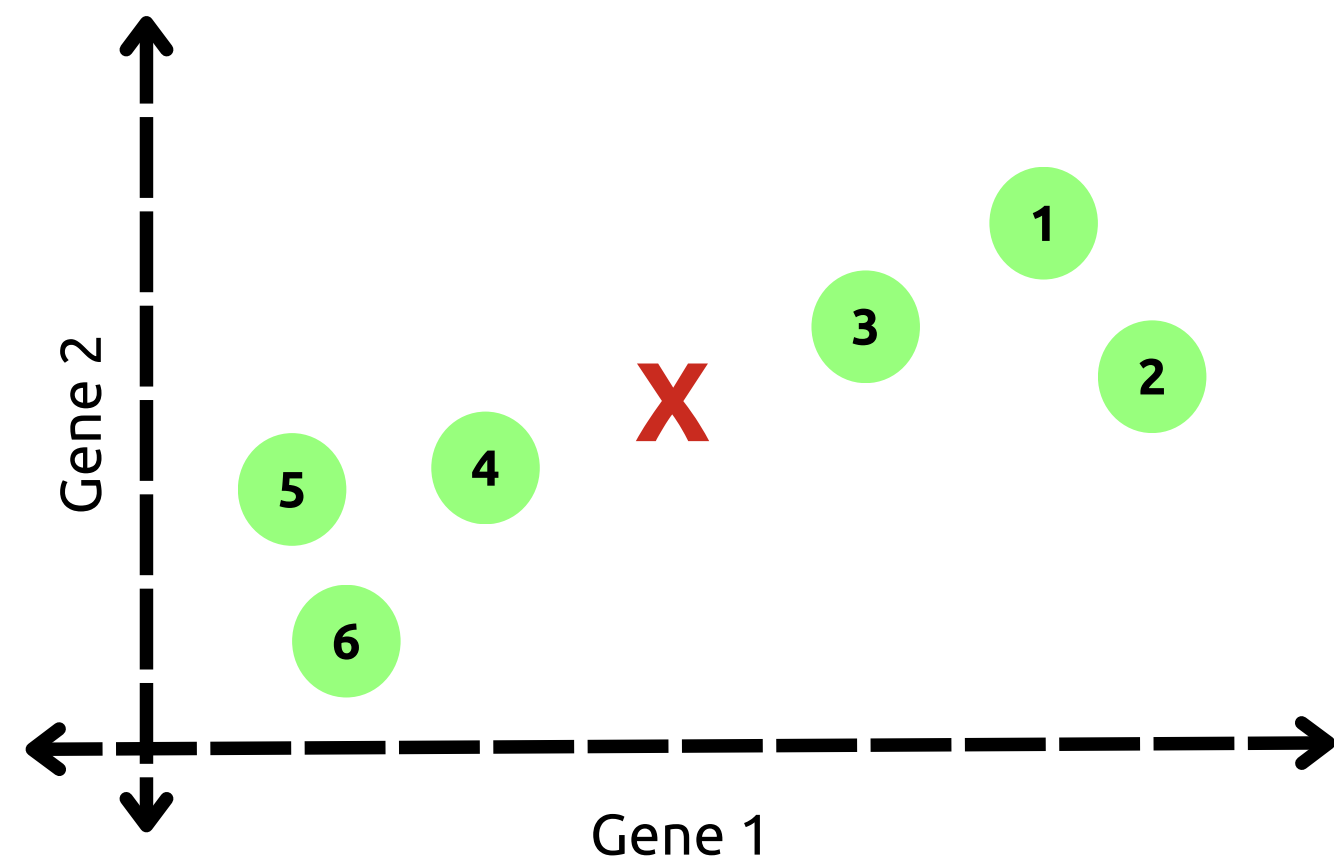
As you can see, iguanas 1, 2 and 3 cluster together, as well as iguanas 4, 5 and 6.



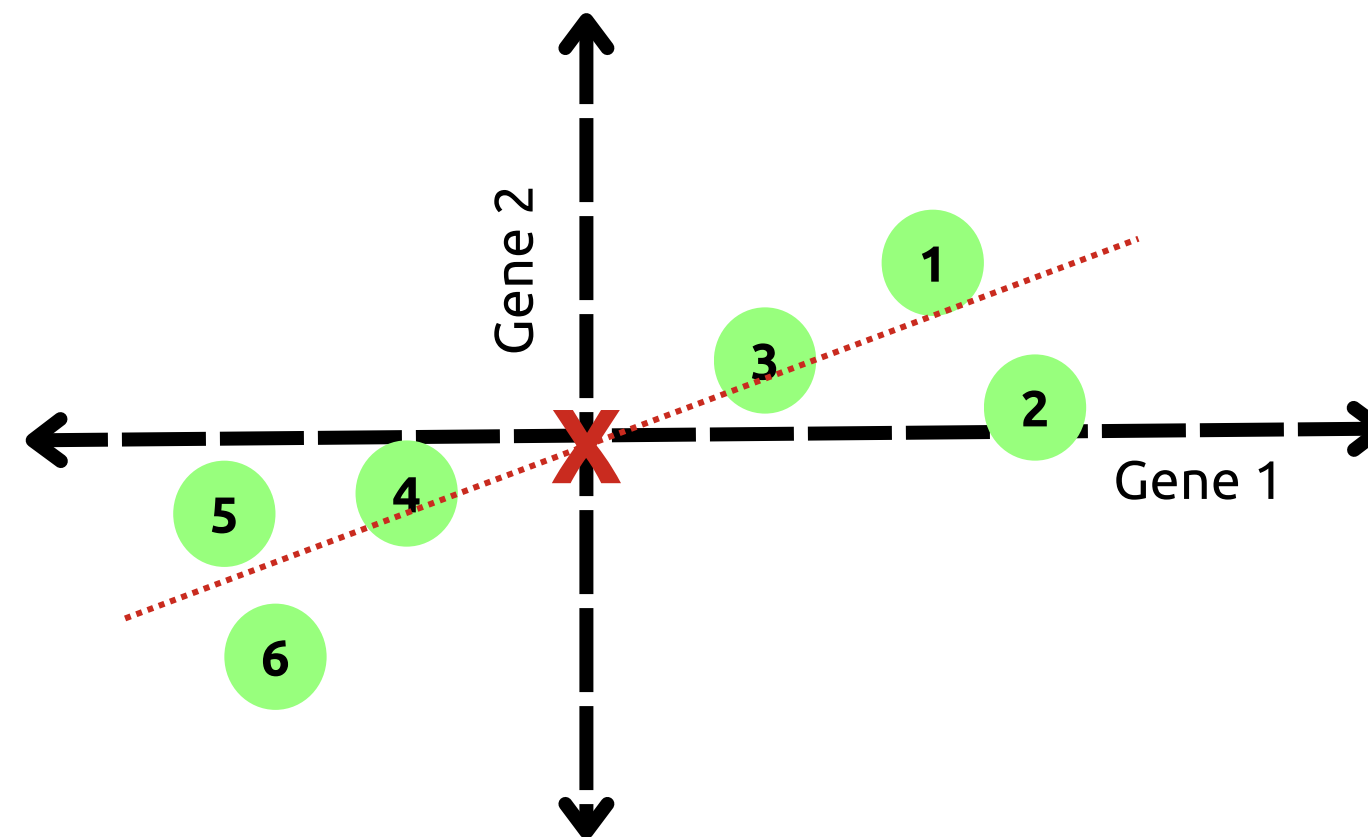
If we analyze three genes, then we would have a 3D plot.

Lets develop the example

PCA is like taking a photograph of a 3D object from the best possible angle to understand its shape.



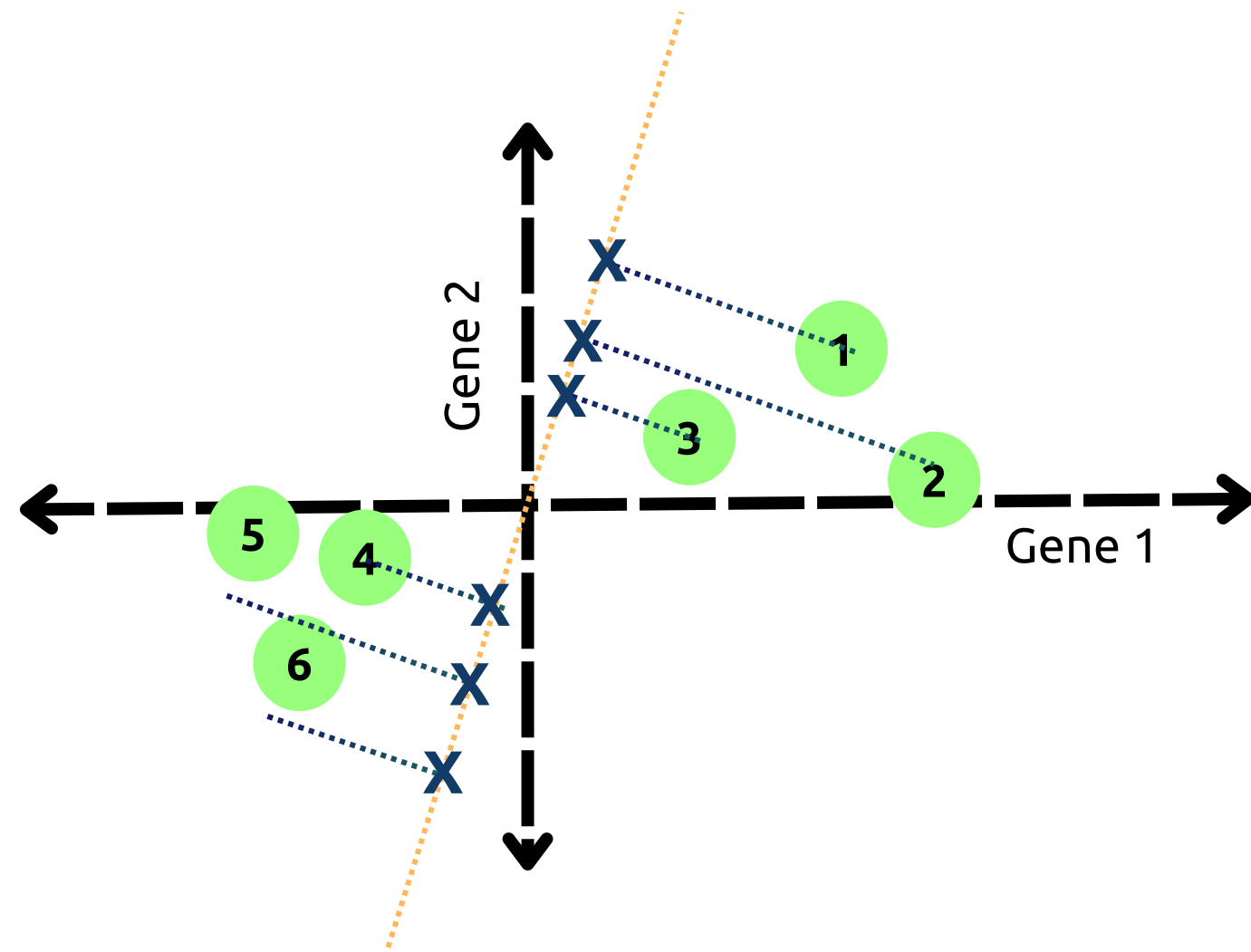
By calculating the average distances of each point to gene 1 and gene 2, we can find the midpoint of the data



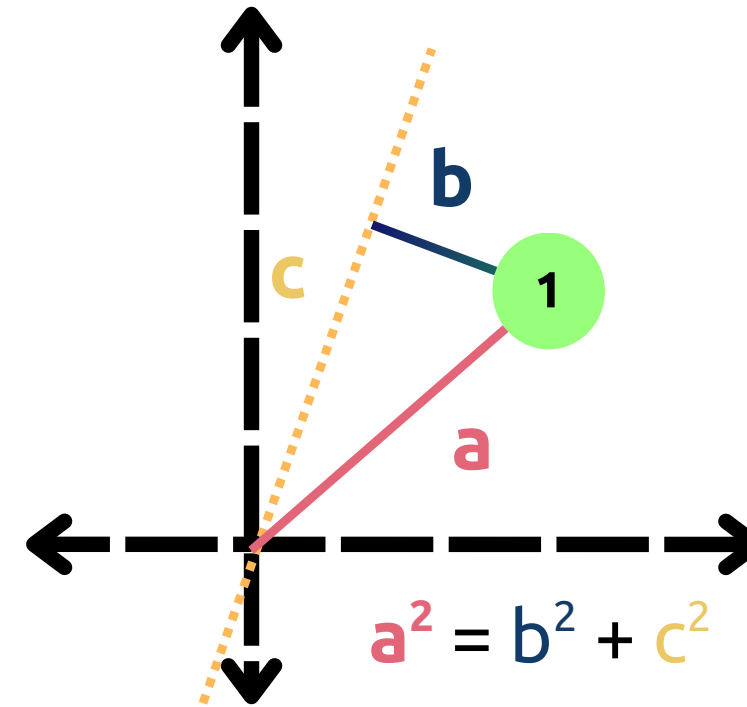
From there, we “draw” a line from the origin that best fits the data

Now we shift the data so the center is in the origin of the graph. Note that this shifting does not change the relative distance between the points.

How do I decide if I have a good fit?



We need to find the line that either minimizes the distances or maximizes the distances from the projected points to the origin. Which is essentially the same line.



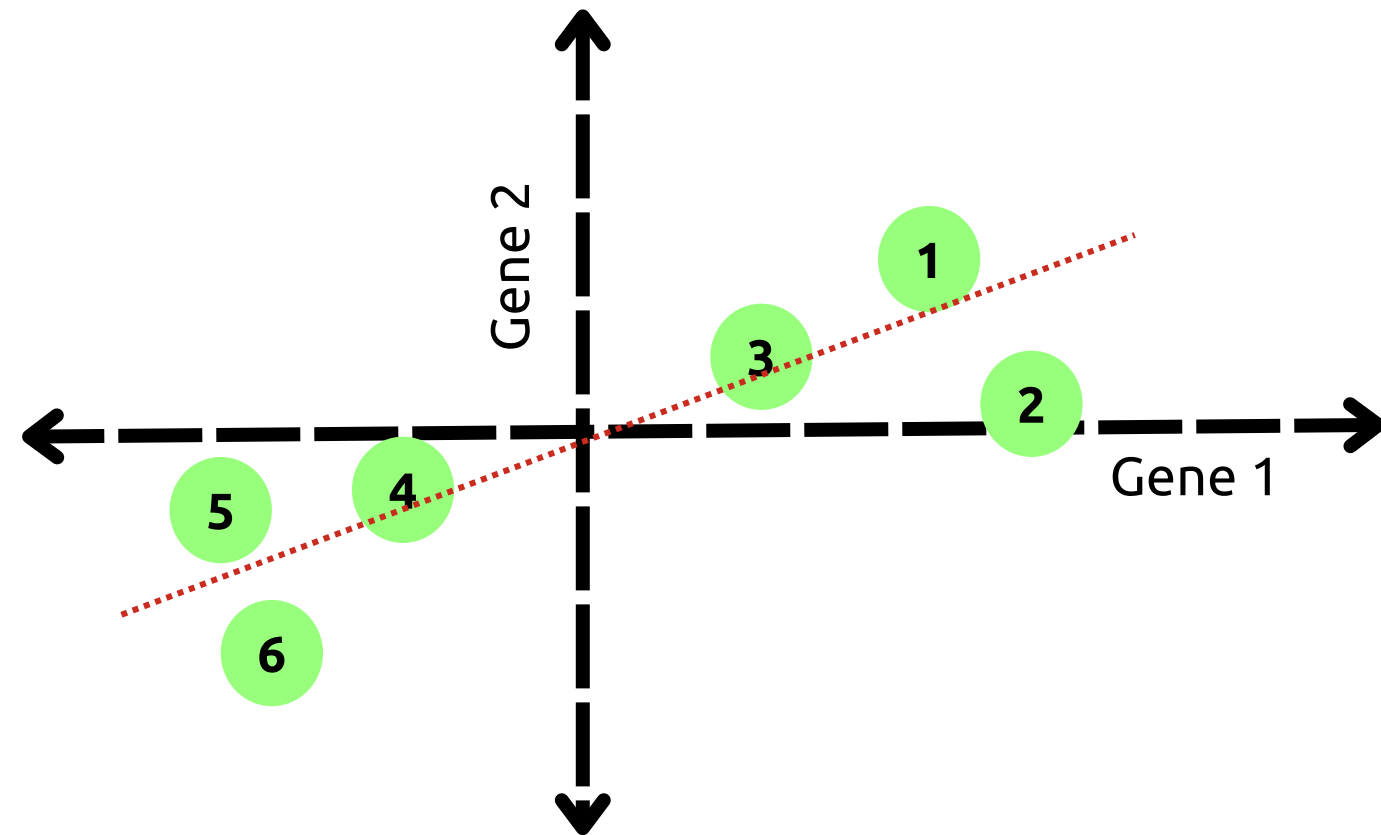
To better visualize this, use the pythagorean theorem. Since the distance from the point to the origin is fixed, **a** will not change (a=hypotenuse).

You can MINIMIZE **b**, or MAXIMIZE **c** to make get a better fit

It is easier to find c, so PCA finds the fitting line by maximizing the squared distances (this way we don't mess with negatives) from the projected points to the origin.

We choose the line with the largest distances AND THAT IS OUR PRINCIPAL COMPONENT!!

The information from the slope



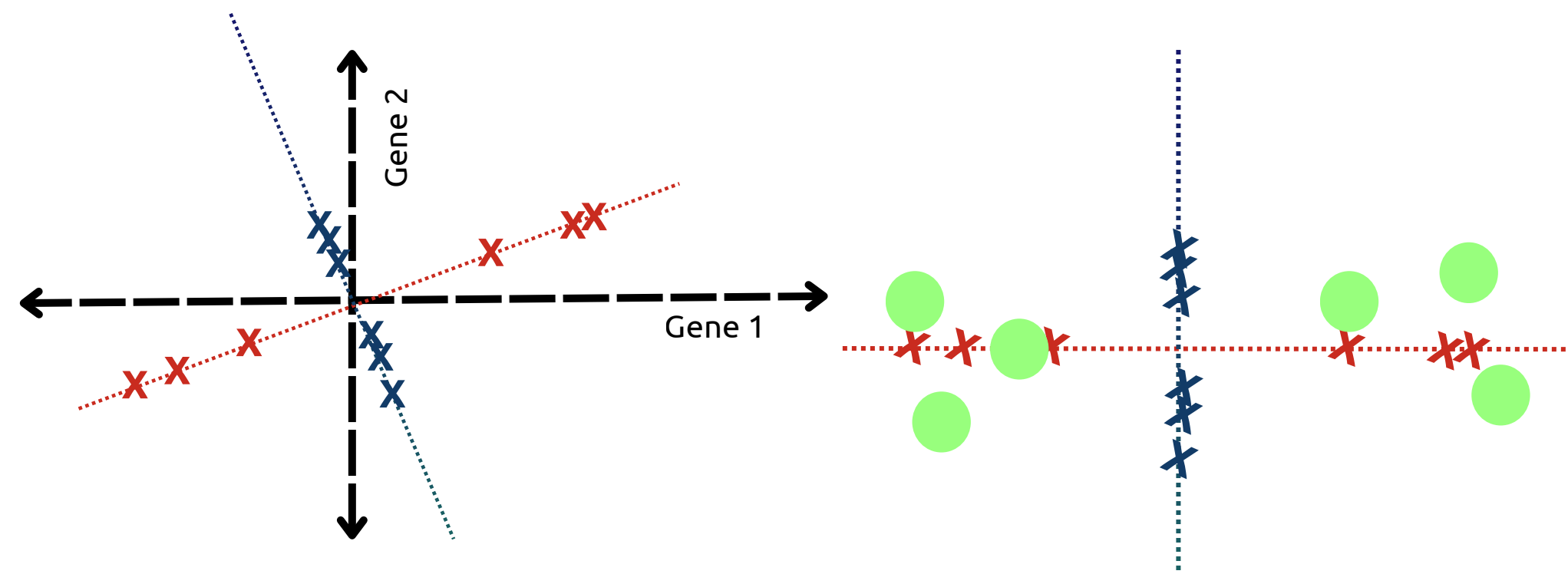
This is our best fit, the slope of the line gives us information about which gene spreads more the data.

In this example, the ratio of gene 1 to gene 2 is higher. So the gene 1 gives us more information of how the data is spread out.

In this 2D example, PC2 is just the perpendicular line to PC1 that goes through the origin. But if you apply the same procedure you should find the exact same fit.

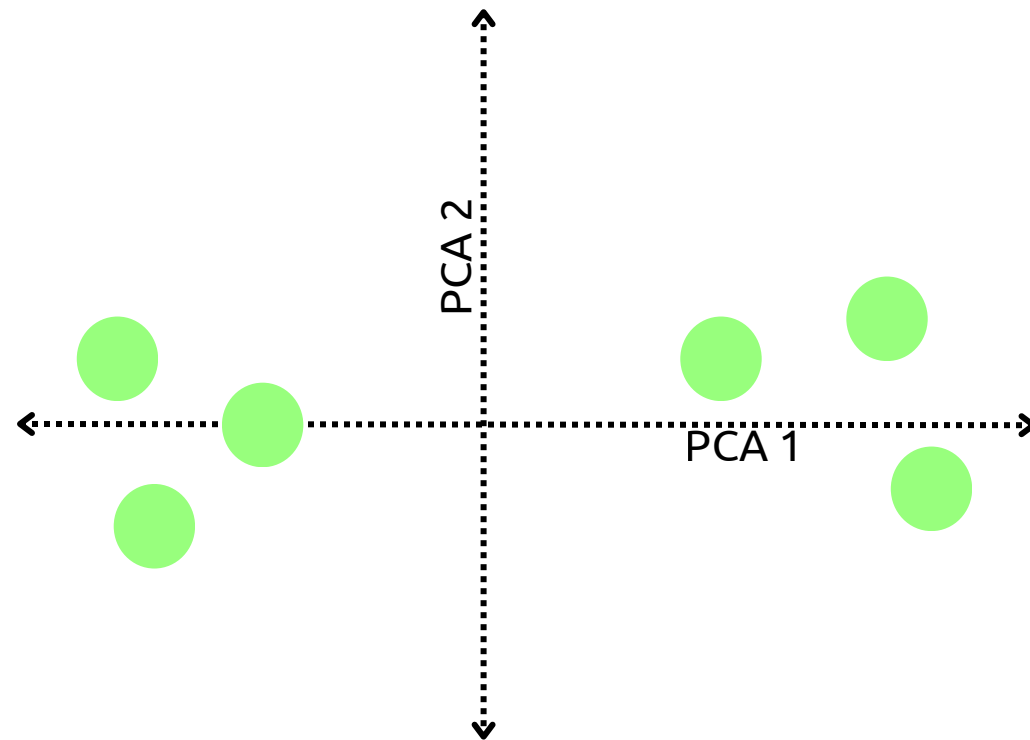
Using PC1 and PC2 you can build a 2D *scree plot*. This way you can visualize the real spread of your iguanas :)

Note we simply made PC1 horizontal and PC2 vertical, and plotted the new points.



Yay! There you have your scree plot

Finalizing your PCA



Remember you summed up the squared distances for each fit line? By dividing these sums by $n-1$ number of data you get *eigenvalues* for each PC.

$$\frac{SS(\text{distances for PC1})}{n - 1} = \text{Variation for PC1}$$

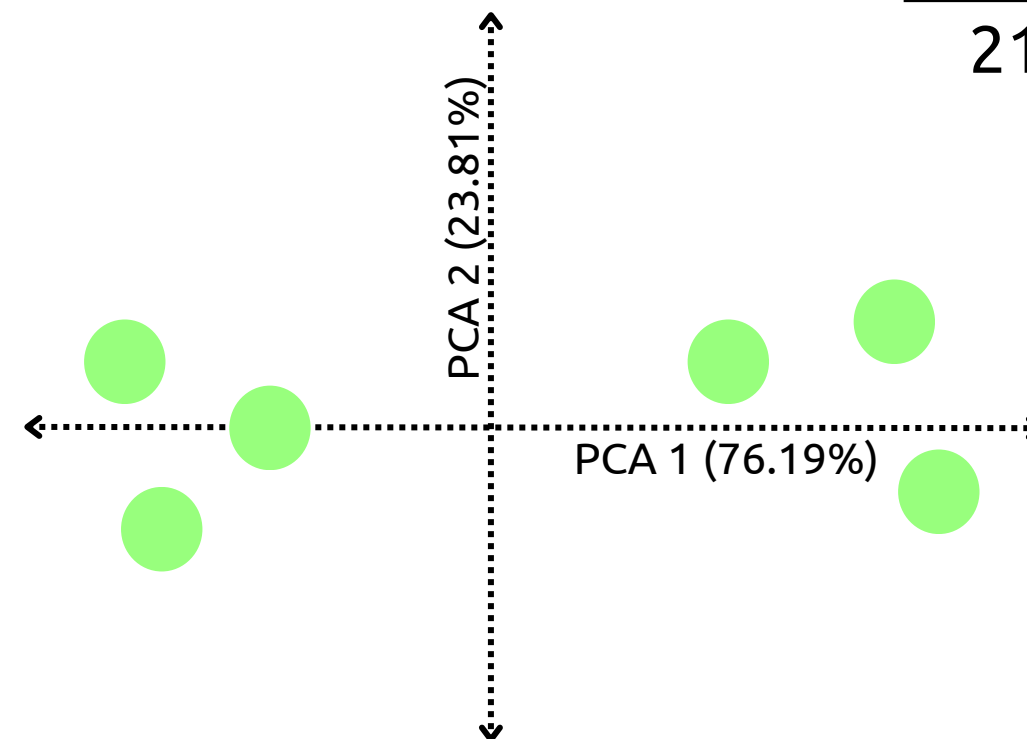
$$\frac{SS(\text{distances for PC2})}{n - 1} = \text{Variation for PC2}$$

If you sum these two variations and find the ratio of each of them with respect to the sum, you get the total variation around the principal components.

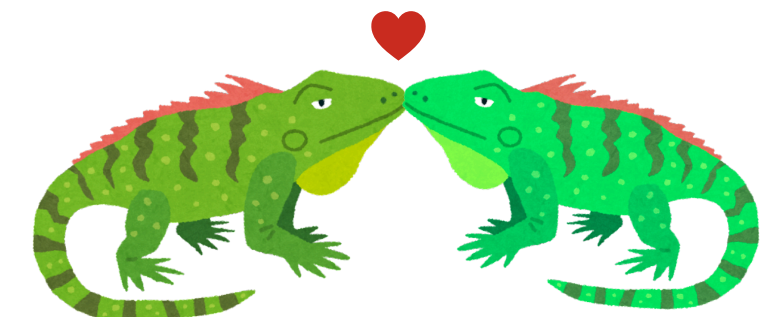
eigenvalue for PC 1 = 16 eigenvalue for PC 2 = 5 summing them = 21

$$\text{Total variation around PC 1} = \frac{16}{21} = 0.76 = 76.19\%$$

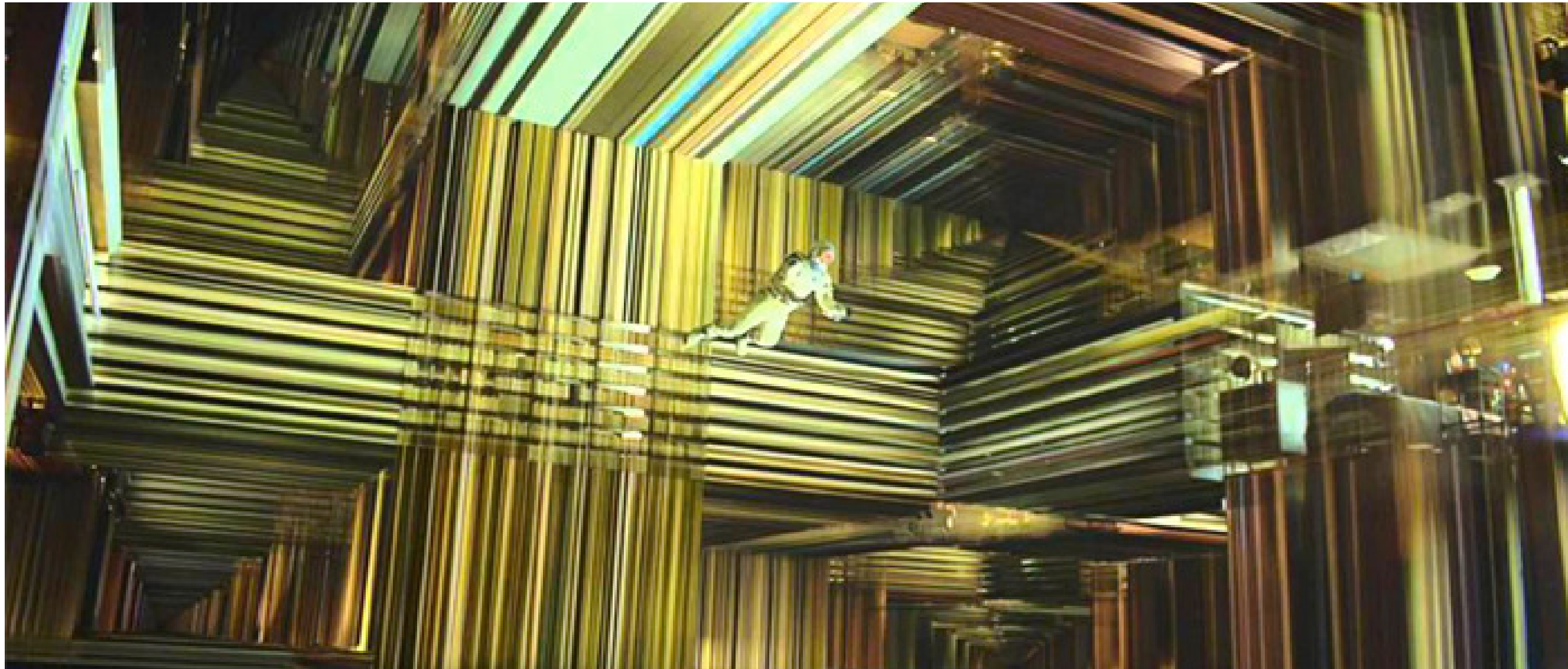
$$\text{Total variation around PC 2} = \frac{5}{21} = 0.23 = 23.81\%$$



And there you have it! You have completed your PCA



And what if I want a hundred genes?



Well... we don't have more dimensions :(

And what if I want a hundred genes?

Biological data is *naturally hyperdimensional*. In this context, each measured variable is a dimension (e.g., each species of bacteria, each gene, or each SNP).

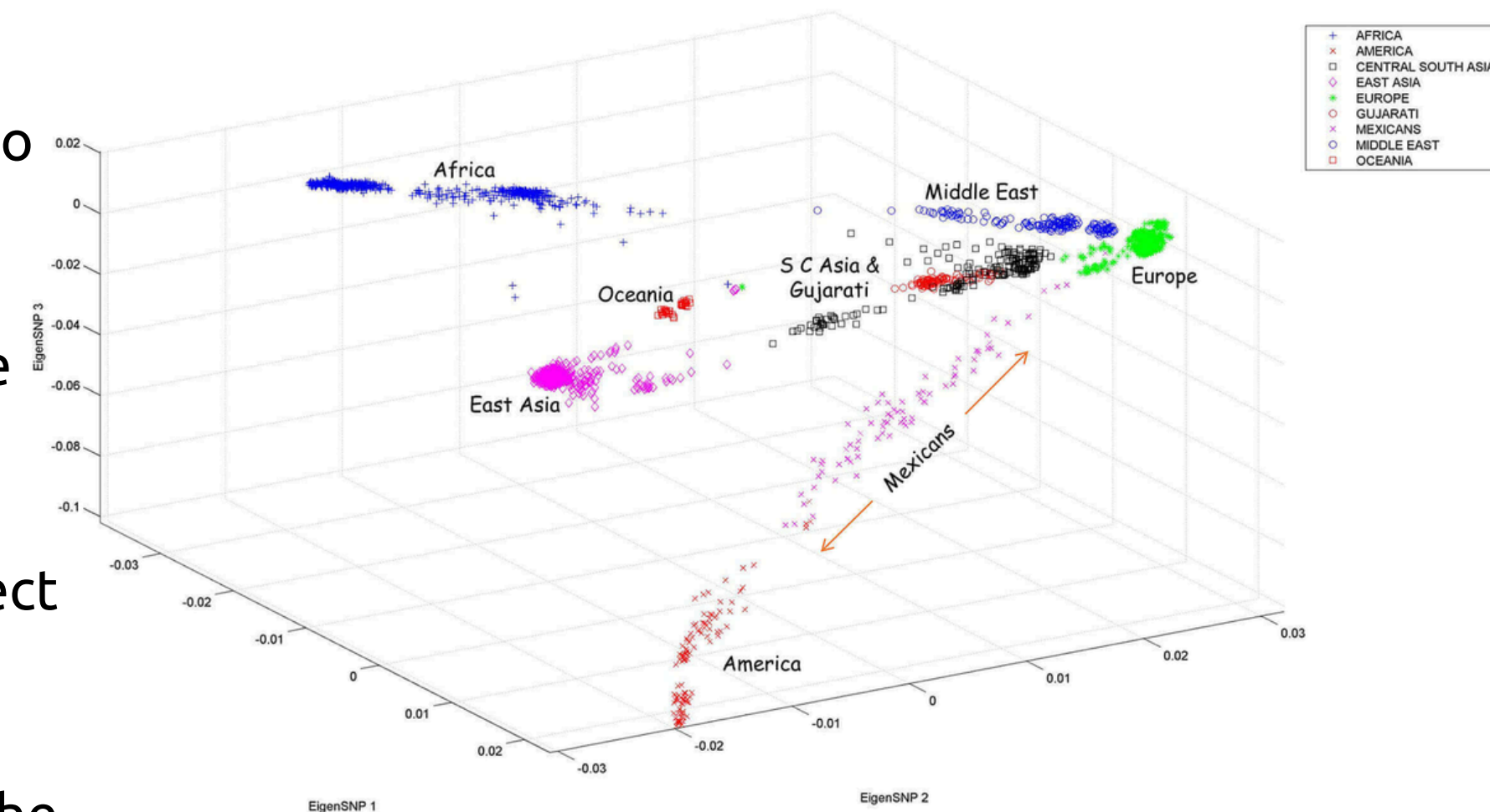
Our visual capacity is limited to 2D or 3D. It is impossible to "see" or graph a table with 1,000 columns of data.

The algorithm "flattens" the N dimension space onto a plane (2D) or cube (3D).

PCA is like moving a camera around a complex object until you find the photo that best describes it.

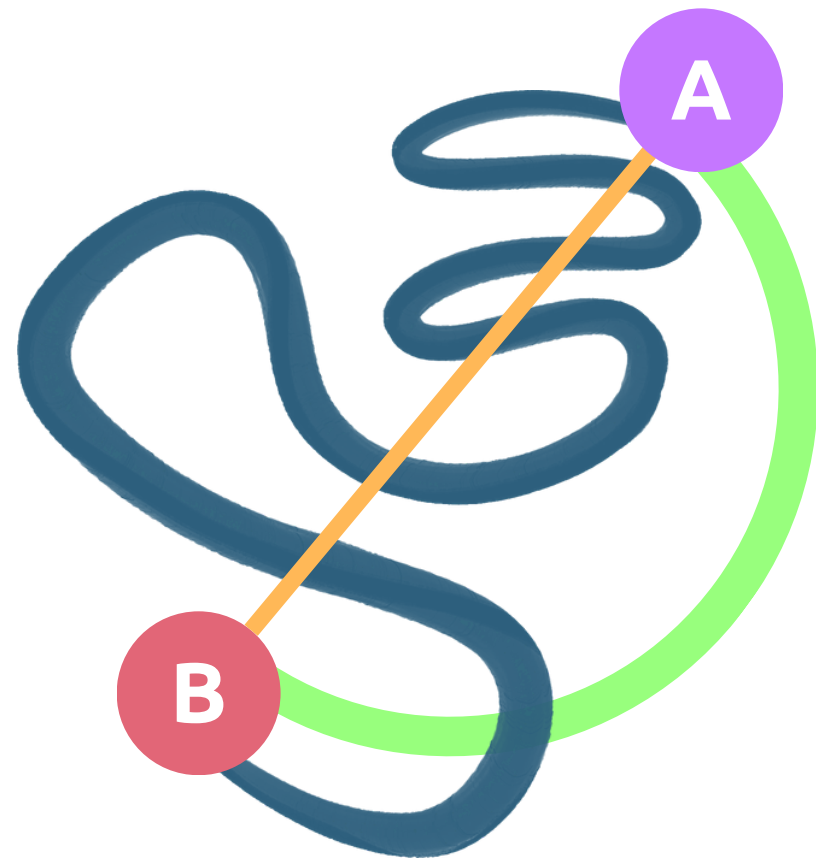
It is not a random elimination of data; the angle where the dispersion of the points is at its maximum is represented in order not to lose relevant information.

PCA reduces dimensionality



MEASURING...

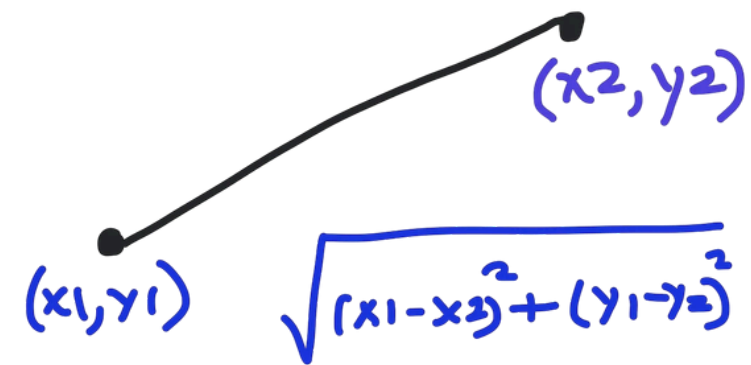
What is the distance between A and B?



Is there an actual *correct* way to measure the distance between them?

The answer is it all depends on the distance that has relevant meaning for you.

Here are some examples of ways to measure distances:



Euclidean Distance

XOR (Exclusive OR) operation

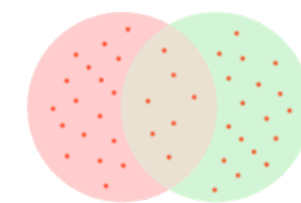
Input A	Input B	XOR output
0	0	0
0	1	1
1	0	1
1	1	0

Hamming Distance = 2

Hamming Distance



Manhattan Distance

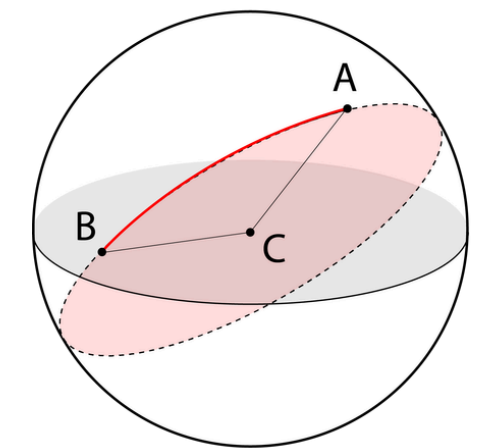


Jaccard similarity

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

total elements in intersection

total elements in union i.e. Universal Set



Great circle distance

Selecting the “best” distance is part of the art of data science -and bioinformatics-

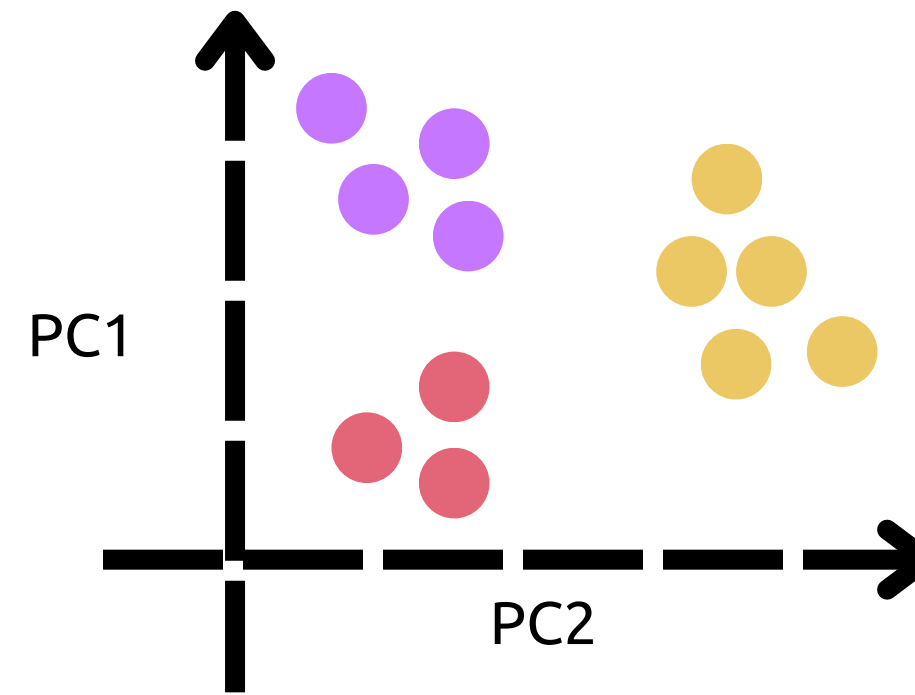
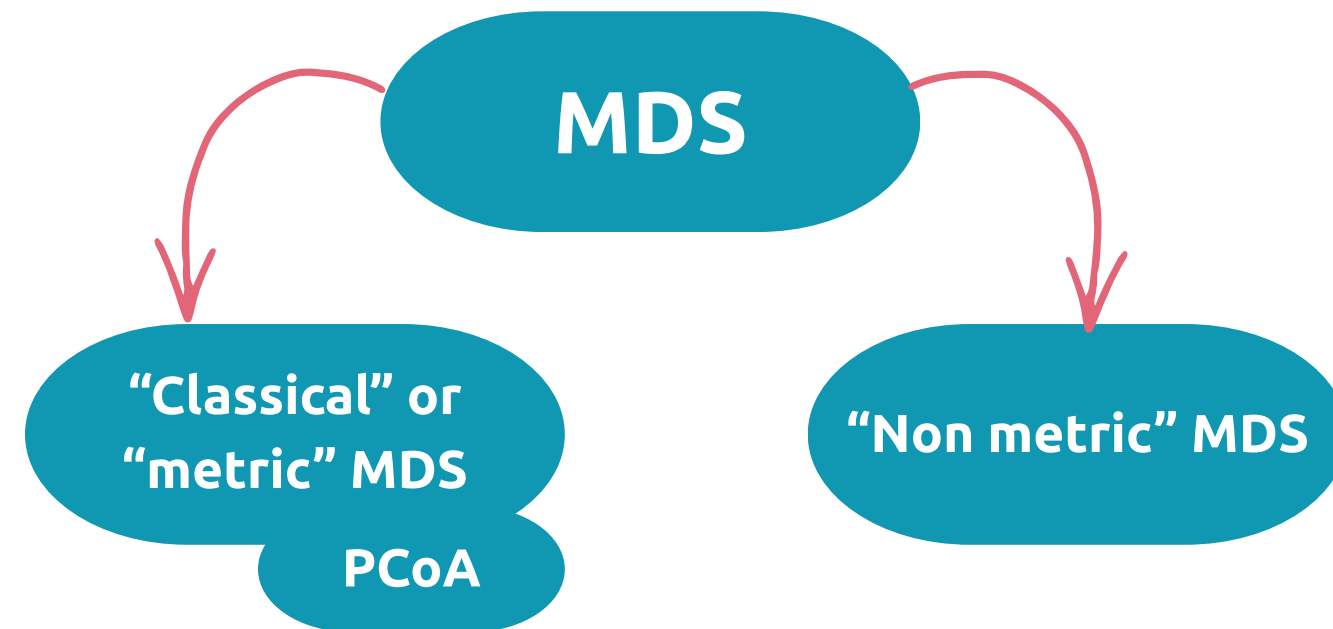
MULTIDIMENSIONAL SCALING (MDS)

PCA and MDS are cousins, not sisters...

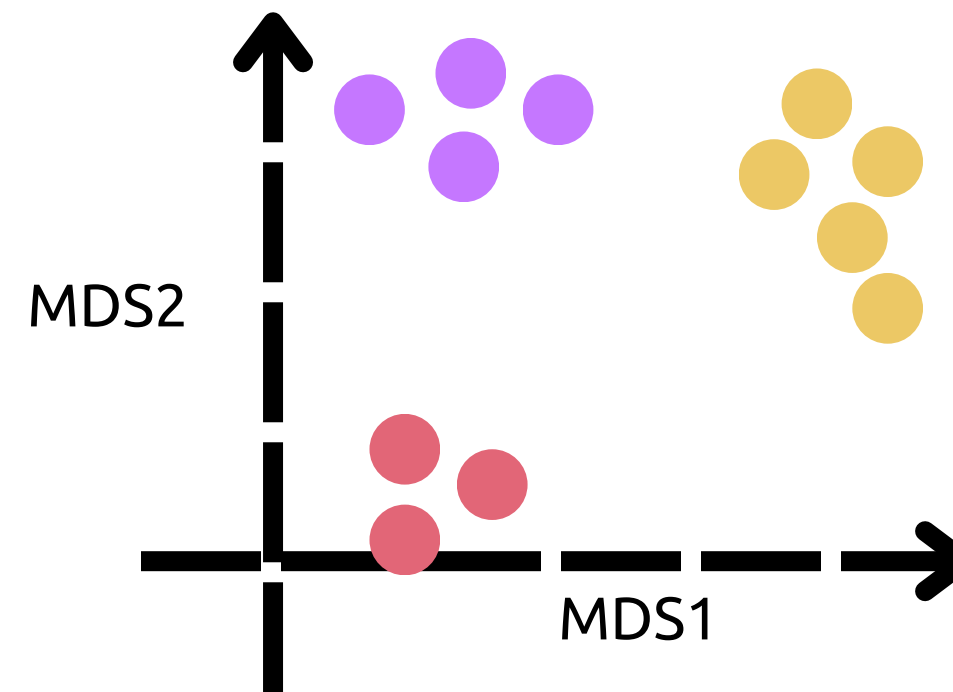
MDS is a *numerical* method for representing the observed similarity among a set of objects.

Thus, we measure distances and assume that the original distances have a precise numerical value and try to preserve them as is.

We have two types of MDS:



PCA creates plots based on **correlations** among samples.



MDS (and PCoA) create plots based on **distances** between samples

PRINCIPAL COORDINATE ANALYSIS (PCOA)

“Classical” multidimensional scaling

Instead of looking at the raw data, PCoA begins by **calculating the distance** between pairs of samples. These results are organized into a distance matrix

	Iguana 1	Iguana 2	Iguana 3
Gene 1	3	1.2	8.1
Gene 2	2.9	4.2	5.4
Gene 3	2.2	11.1	9.1
Gene 4	3.5	1.9	7.4
Gene 5	4.2	1.5	2.3
Gene 6	2.8	1.4	3.6

The power of PCoA lies in the fact that you choose the distance metric according to your biological problem.

For example:

- If you measure the distances with the euclidean method, you’d get the exact same result as a PCA (*why does that happen?*)
- If you're interested in seeing how much gene expression changes on a logarithmic scale you could use a log fold change

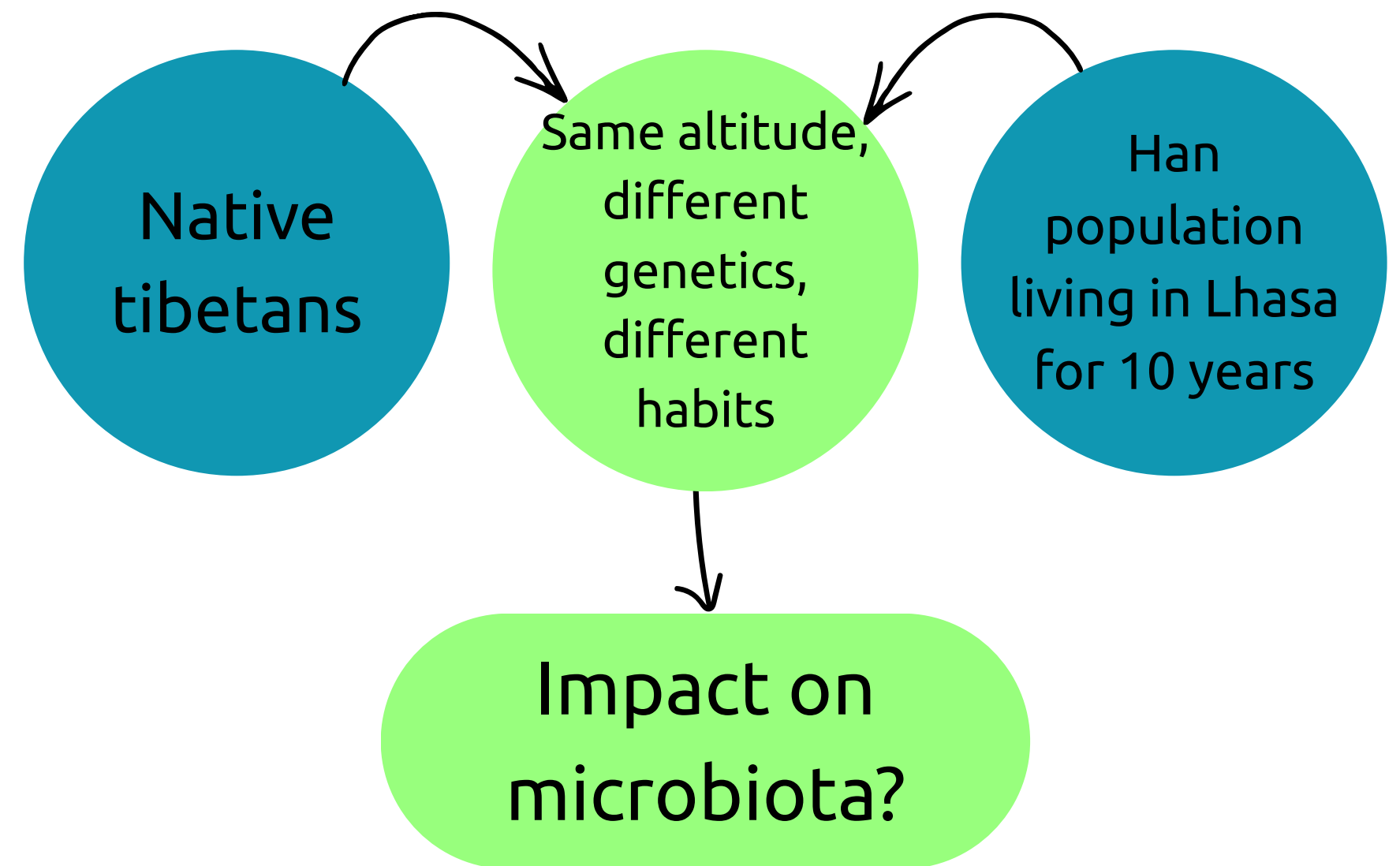
VISUALIZE HOW WE USE IT

A real life application of these concepts

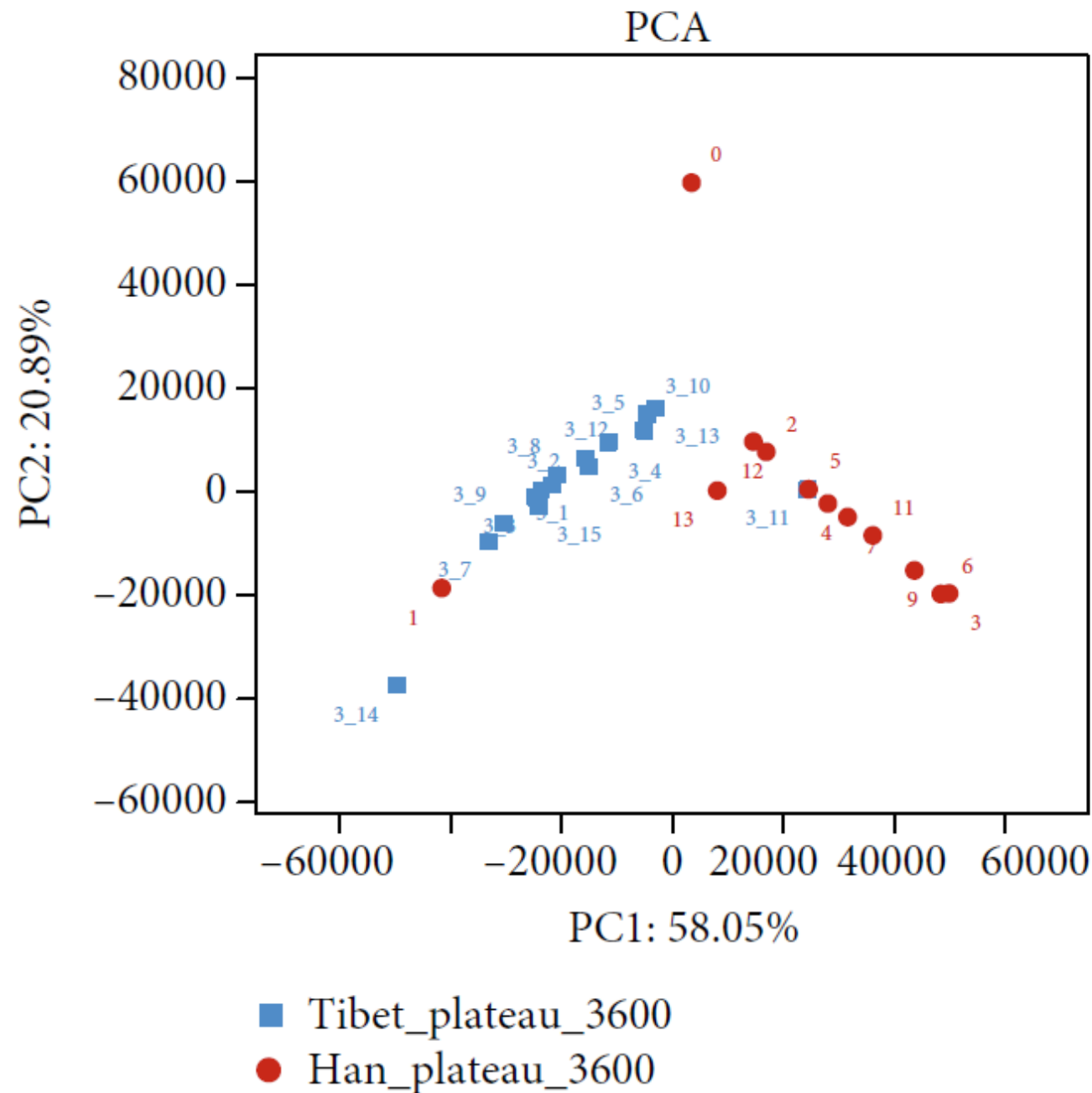


Research Article

Host Genetic and Environmental Factors Shape the Composition and Function of Gut Microbiota in Populations Living at High Altitude



See how they used these two...



(c)

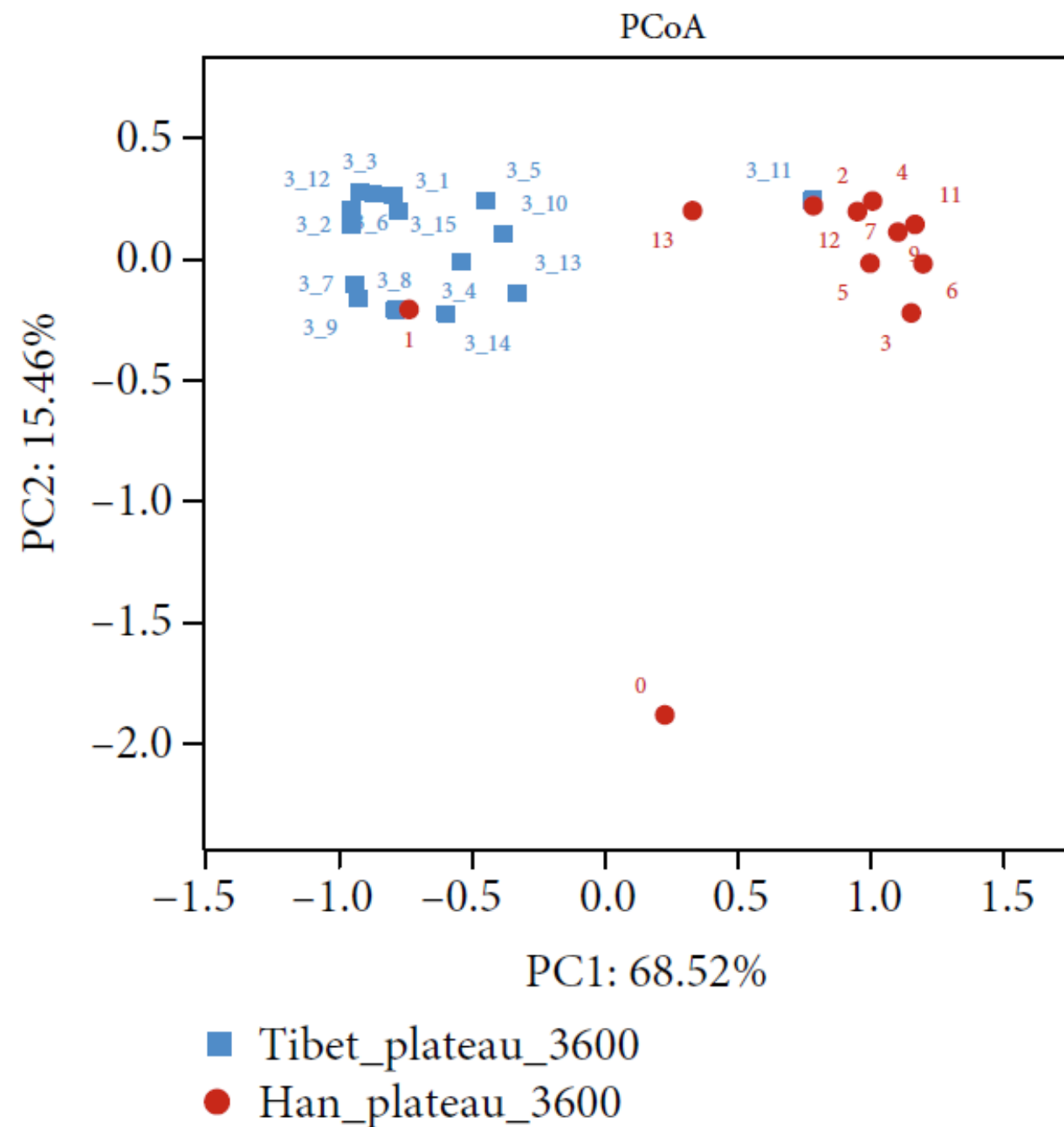
This PCA is showing differences in the gut microbiota between the native Tibetans and the Hans based on the **composition** of the bacterial communities.

The blue dots (Native tibetans) are mostly on the left

The red dots (Han immigrants) are mostly on the right

There is a clear separation. This indicates that the **bacterial "signature"** of a Tibetan is intrinsically different from that of a Han migrant, based simply on the proportions of the bacteria present.

See how they used these two...



(d)

This PCoA is showing differences in the gut microbiota between the native Tibetans and the Hans based on the **distance matrix of Bray-Curtis dissimilarity** of the microbial community.

Unlike PCA, PCoA does not use raw data directly, but first calculates a **distance matrix** using the Bray-Curtis metric.

It not only checks if a phylum is present or not, but also ignores "double zeros" (if both are missing a phylum, it doesn't count as a similarity).

The separation is even more pronounced and "clean" than in the PCA. This confirms that the difference between the microbial communities is robust and not a statistical artifact.

THANKS!!!!